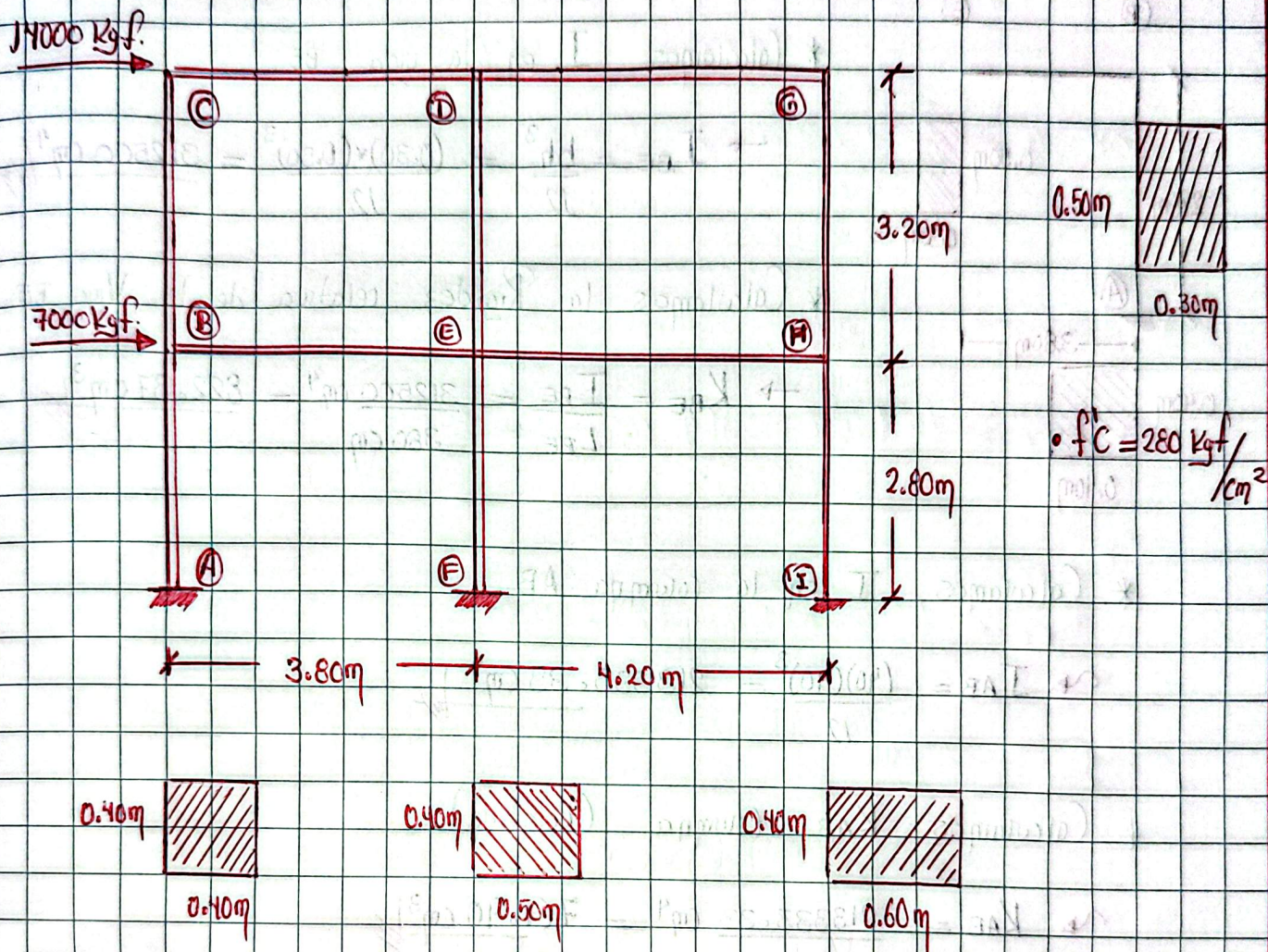


Ejercicio: Resolver el portico usando el Método de Muto



Solución:

1° Determinamos el modulo de elasticidad.

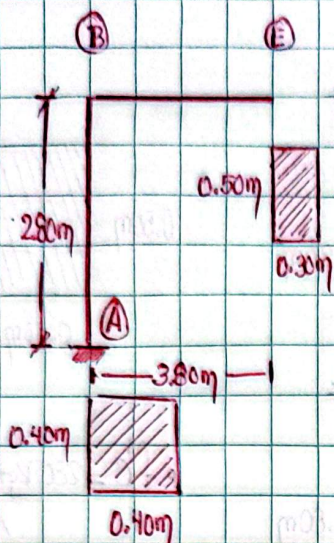
Sabemos que ... $f'_c = 280 \text{ kgf/cm}^2$

$$\rightarrow E = 15100 \sqrt{f'_c}$$

$$\rightarrow E = 15100 \sqrt{280}$$

$$E = 252671.33 \text{ kgf/cm}^2$$

2º Determinamos la rigidez de las columnas



* Calculamos I en la viga BE

$$\rightarrow I_{BE} = \frac{bh^3}{12} = \frac{(0.30) \times (0.50)^3}{12} = \underline{312500 \text{ cm}^4}$$

* Calculamos la Rigidez relativa de la Viga BE

$$\rightarrow K_{BE} = \frac{I_{BE}}{L_{BE}} = \frac{312500 \text{ cm}^4}{380 \text{ cm}} = \underline{822.37 \text{ cm}^3}$$

* Calculamos I en la columna AB

$$\rightarrow I_{AB} = \frac{(40)(40)^3}{12} = \underline{213333.33 \text{ cm}^4}$$

* Calculamos K_{AB} columna (Relativa)

$$\rightarrow K_{AB} = \frac{213333.33 \text{ cm}^4}{280 \text{ cm}} = \underline{761.90 \text{ cm}^3}$$

* Calculamos la rigidez de la columna AB

$$\rightarrow \bar{K}_{AB} = \frac{K_{BE}}{K_{AB}} = \frac{822.37 \text{ cm}^3}{761.90 \text{ cm}^3} = \underline{1.08}$$

Ahora

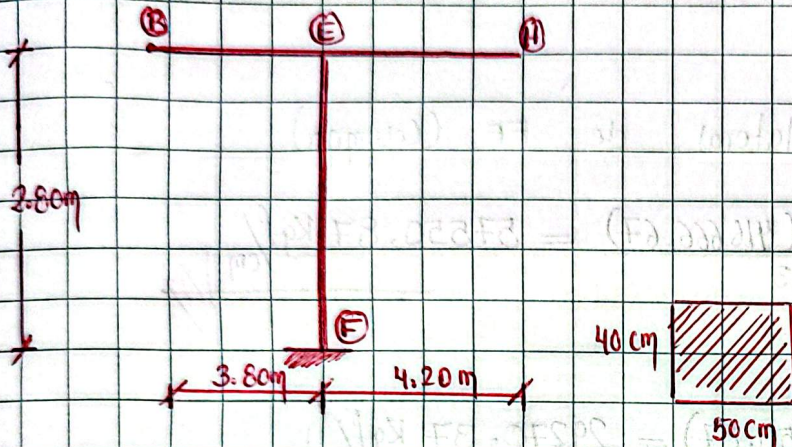
$$\rightarrow a = \frac{0.5 + \bar{K}_{AB}}{2 + \bar{K}_{AB}} = \frac{0.5 + 1.08}{2 + 1.08}$$

$$a = \underline{0.5129}$$

* Calculamos la rigidez lateral de la columna AB

$$\rightarrow K_{AB}^0 = \frac{12 \times E \times I}{h^3} = \frac{12 \times (252671.33) \times (213333.33)}{(280)^3} = 29466.04 \text{ Kg/cm}^2$$

$$\rightarrow K_{AB}^{PE} = 0.5 \times K_{AB}^0 = (0.5) \times (29466.04) = 14733.02 \text{ Kg/cm}^2$$



* Calculamos I_{BE} y I_{EH} (Viga)

$$\rightarrow I_{BE} = I_{EH} = 312500 \text{ cm}^4$$

* Calculamos la rigidez relativa de BE y EH (Viga)

$$* K_{BE} = 822.37 \text{ cm}^3$$

$$* K_{EH} = 744.05 \text{ cm}^3$$

* Calculamos I_{EF} (Columna)

$$* I_{EF} = \frac{(40)(50)^3}{12} = 416666.67 \text{ cm}^4$$

* Calculamos la rigidez relativa EF (Columna)

$$\rightarrow K_{EF} = \frac{416666.67 \text{ cm}^4}{280 \text{ cm}} = 1488.09 \text{ cm}^3$$

* Calculamos la rigidez de la columna FE

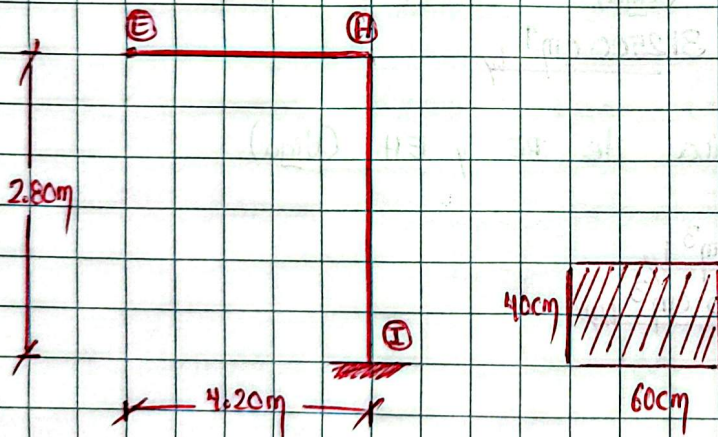
$$\rightarrow \bar{K}_{FE} = \frac{K_{ED} + K_{EH}}{K_{FE}} = \frac{822.37 + 744.05}{1488.09} = 1.0526$$

$$\rightarrow \alpha = \frac{0.5 + 1.0526}{2 + 1.0526} = 0.5086$$

* Calculamos la rigidez lateral de FE (Columna)

$$\rightarrow K_{FE}^{\circ} = \frac{(12)(252671.33)(416666.67)}{(280)^3} = 57550.87 \text{ Kg/cm}$$

$$K_{FE}^{PE} = (0.5086) \times (57550.87) = 29270.37 \text{ Kg/cm}$$



* $I_{EH} = 312500 \text{ cm}^4$

* $K_{EH} = 744.05 \text{ cm}^3$

* Calculamos I_{HI} (Columna)

$$\rightarrow I_{HI} = \frac{(40) \times (60)^3}{12} = 720000 \text{ cm}^4$$

* Calculamos la K_{HI} (columna)

$$\hookrightarrow K_{HI} = \frac{I_{HI}}{L_{HI}} = \frac{720000 \text{ cm}^4}{280 \text{ cm}} = 2571.43 \text{ cm}^3$$

* Calculamos la $\bar{K}_{HI}$ (columna)

$$\rightarrow \bar{K}_{HI} = \frac{744.05}{2571.43} = 0.2894$$

$$\hookrightarrow \alpha = \frac{0.5 + 0.2894}{2 + 0.2894} = 0.3448$$

* Calculamos la rigidez lateral de la columna (HI)

$$\rightarrow K_{HI}^{\circ} = \frac{(12) \times (252671.33) \times (720000)}{(280)^3} = 99447.89 \text{ Kg/cm}$$

$$\hookrightarrow K_{HI}^{PE} = (0.3448) \times (99447.89) = 34289.63 \text{ Kg/cm}$$

14000 Kg

